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RFC-ATF-6

Agent Trust Fabric — Behavioral Execution Verification Protocol

Behavioral Anchor Record · Constraint Conformance Signal · Cross-Turn Coherence Hash Chain

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Version 1.0.0 · May 2026 · DOI: pending submission

Extension to RFC-ATF-1, RFC-ATF-2, RFC-ATF-3, RFC-ATF-4, and RFC-ATF-5

Compliance Designation: ATF-BEV-Compliant — Sixth and highest tier

18 new invariants (BEV-INV-001–018) · 106 total ATF invariants across 20 protocol families

Abstract

RFC-ATF-6 specifies the **Behavioral Execution Verification Layer (BEV)** of the Agent Trust Fabric — the sixth and final RFC in the ATF Open Standard series. RFC-ATF-6 answers the structural question that no prior RFC addresses: *did the agent actually behave within its authorized constraints during execution, turn by turn?*

The five prior RFCs answered questions about governance infrastructure: who authorized (RFC-ATF-1), was authority continuously valid (RFC-ATF-2), where does evidence go (RFC-ATF-3), was monitoring active between requests (RFC-ATF-4), what else could have happened and is compliance universal (RFC-ATF-5). All five share a structural assumption: the authorized action is treated as a black box. The governance receipt authorizes. The execution receipt records. But no ATF artifact through RFC-ATF-5 contains a cryptographic record of what the agent actually produced — the Behavioral Execution Gap (Gap_BEV).

RFC-ATF-6 closes this gap with three new protocol components: the **Behavioral Anchor Record (BAR)**, the **Constraint Conformance Signal (CCS)**, and the **Cross-Turn Coherence Hash Chain (CTCHC)**. These components are additive, backward compatible, and constitute Layer 6 of the ATF stack.

Figure 1 — ATF 6-Layer Protocol Stack

The Behavioral Execution Verification Layer (BEV) occupies Layer 6 — the topmost layer of the ATF stack. Each layer is additive: BEV does not replace or modify any prior layer. Together the six layers cover the complete lifecycle of a governed autonomous agent action.

ATF Protocol Stack — 6 Layers · 106 Invariants

Layer 1 RFC-ATF-1	**Identity & Delegation** AIR · DR	6 inv.
Layer 2 RFC-ATF-2	**Runtime Continuity** TAR · RCR · AFG	14 inv.
Layer 3 RFC-ATF-3	**Evidence Lifecycle** OEP · GPIL · FEA	40 inv.
Layer 4 RFC-ATF-4	**Proactive Governance** AGVP · DSPP · SSD	70 inv.
Layer 5 RFC-ATF-5	**Cognitive Governance** CGE · GUGT · TGB	88 inv.
Layer 6 RFC-ATF-6	**Behavioral Exec. Verif.** BAR · CCS · CTCHC	106 inv.

Figure 1: ATF protocol stack showing all 6 RFC layers. Layer 6 (BEV, gold) adds behavioral attestation atop the five existing layers. Each layer's invariant count is cumulative.

1. Problem Statement — The Behavioral Execution Gap

1.1 Gap_BAG — Behavioral Attestation Gap

Every prior ATF artifact treats the authorized action as a black box. The Governing Receipt (GR) authorizes the action. The Execution Receipt records that the action occurred. But no artifact cryptographically binds the GR to what the agent actually produced during execution — the Behavioral Output (BO). This is Gap_BAG:

```
Gap_BAG = { (GR, BO) : no verifiable binding exists between GR and BO }

Consequences:
a) Regulatory gap: EU AI Act Art. 9 requires monitoring that AI operates within defined boundaries
b) Forensic gap: auditor cannot reconstruct what the agent actually said or did
c) Liability gap: legal counsel cannot prove agent operated within authorized constraints
```

1.2 Gap_COP — Conformance Observability Problem

The AGVP watchdog (RFC-ATF-4) issues anticipatory vetoes based on observable conformance signals. For structural governance (CES, fragmentation) and semantic drift (DSPP), signals exist. For behavioral output conformance, no signal existed before this RFC. Gap_COP left the AGVP input space incomplete for behavioral trajectories.

1.3 Gap_MCP — Multi-Turn Coherence Problem

A set of valid BAR records does not prove Session Coherence without a chaining mechanism. An adversary could present a subset of BARs (omitting turns where violations occurred), reorder BARs, or substitute BARs from a different session. Gap_MCP is the absence of a cryptographic Session Integrity Proof.

Figure 2 — Three Behavioral Gaps Closed by BEV

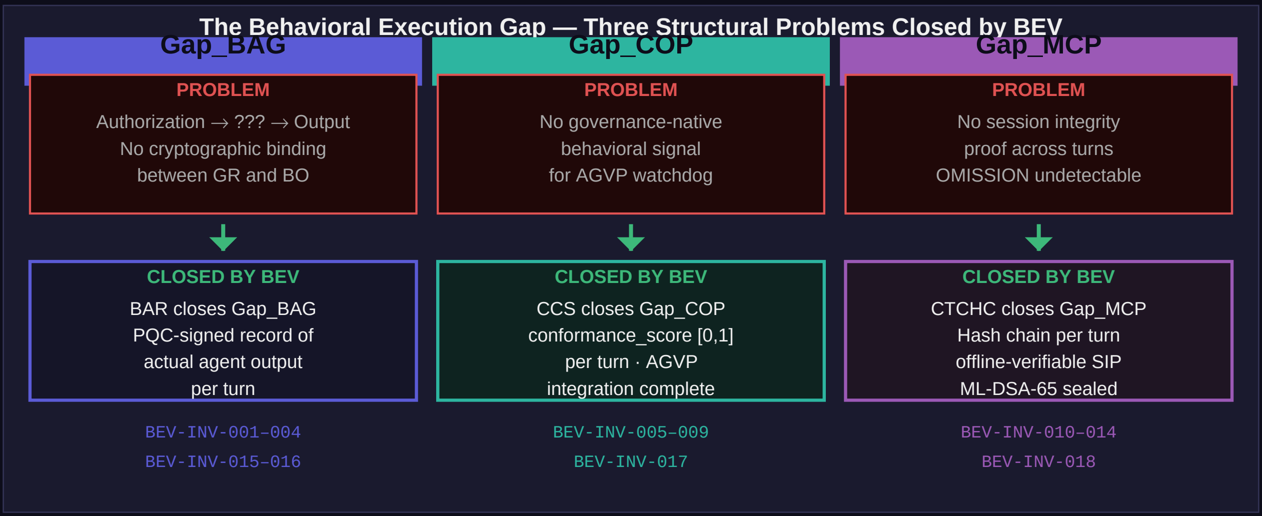

Figure 2: The three structural problems RFC-ATF-6 closes. Each gap is mapped to the BEV component that resolves it: BAR closes Gap_BAG, CCS closes Gap_COP, CTCHC closes Gap_MCP.

2. Behavioral Anchor Record (BAR)

The BAR is a PQC-signed record produced at each execution turn, binding the SHA3-256 hash of the agent's output to the governing receipt and turn index. It is the first ATF artifact that closes the authorization-to-output chain.

2.1 Content Hash Construction

```
output_hash = SHA3-256(output_text.encode('utf-8'))
content_hash = SHA3-256(output_hash || governing_receipt_id || str(turn_index))
pqc_signature = dilithium3.sign(content_hash.encode('utf-8'), platform_secret_key)
pqc_algorithm = 'ML-DSA-65' # FIPS 204
```

2.2 BAR Architecture Flow

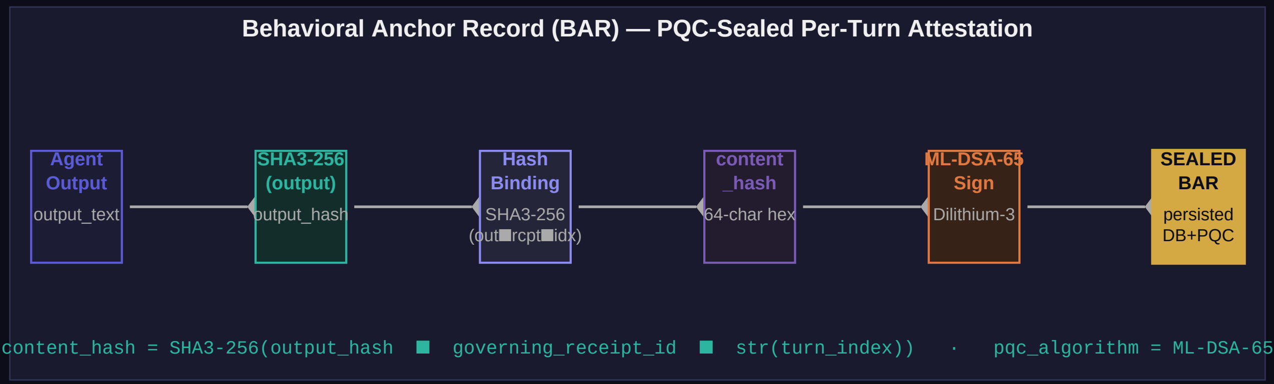

Figure 3: BAR creation pipeline. Every turn's output is hashed, bound to the governing receipt and turn index, then PQC-signed before the BAR is persisted. The three-input binding makes output, receipt, and position simultaneously tamper-evident.

2.3 Output Hash Privacy Modes

Mode	output_payload stored?	CCS computable?	Use case
FULL	Yes — full output	Yes	Forensic / internal audit
HASHED	No — hash only	Partial	Production default (GDPR)
REDACTED	No — omitted	No	Privacy-sensitive regulated

2.4 BAR Invariants

BEV-INV-001	**Mandatory BAR Production**
Statement:	Every execution turn MUST produce a BAR before agent output is delivered. No path delivers output without a BAR.
Formal:	`EXISTS(BAR b) WHERE b.turn_index = i AND b.created_before_output_delivery`

BEV-INV-002	**Content Hash Binding**
Statement:	content_hash = SHA3-256(output_hash || governing_receipt_id || str(turn_index)). All three inputs are simultaneously tamper-evident.
Formal:	`b.content_hash = SHA3-256(b.output_hash || b.governing_receipt_id || str(b.turn_index))`

BEV-INV-003	**HALTED BAR Immediately Halts Session**
Statement:	A BAR with bar_status=HALTED MUST immediately transition the session to SESSION_HALTED. No subsequent turn permitted.
Formal:	`BAR(T_i).bar_status = HALTED IMPLIES session_status(S) = STATUS_HALTED`

BEV-INV-004	**BAR PQC-Sealing and Offline Verifiability**
Statement:	Every BAR MUST be sealed with ML-DSA-65 over its content_hash. No UPDATE or DELETE on atf_behavioral_anchor_records.
Formal:	`dilithium3.verify(b.content_hash.encode(), b.pqc_signature, platform_public_key) = True`

BEV-INV-015	**Empty Output Text Is a Violation**
Statement:	output_text = '' or whitespace-only MUST set bar_status = VIOLATION. Silent outputs are not permissible.
Formal:	`output_text(b).strip() = '' IMPLIES b.bar_status = VIOLATION`

BEV-INV-016	**BAR Identifier Format**
Statement:	Every BAR id MUST conform to 'BAR-{HEX16}' — exactly 16 uppercase hexadecimal characters.
Formal:	`MATCHES(b.bar_id, r'^BAR-[0-9A-F]{16}$')`

3. Constraint Conformance Signal (CCS)

The CCS provides a governance-native behavioral conformance measurement for each execution turn. 'Governance-native' means it references the actual Constraint Vector (CV) from the Governing Receipt — the receipt IS the measurement specification, PQC-signed and immutable.

3.1 Score Components

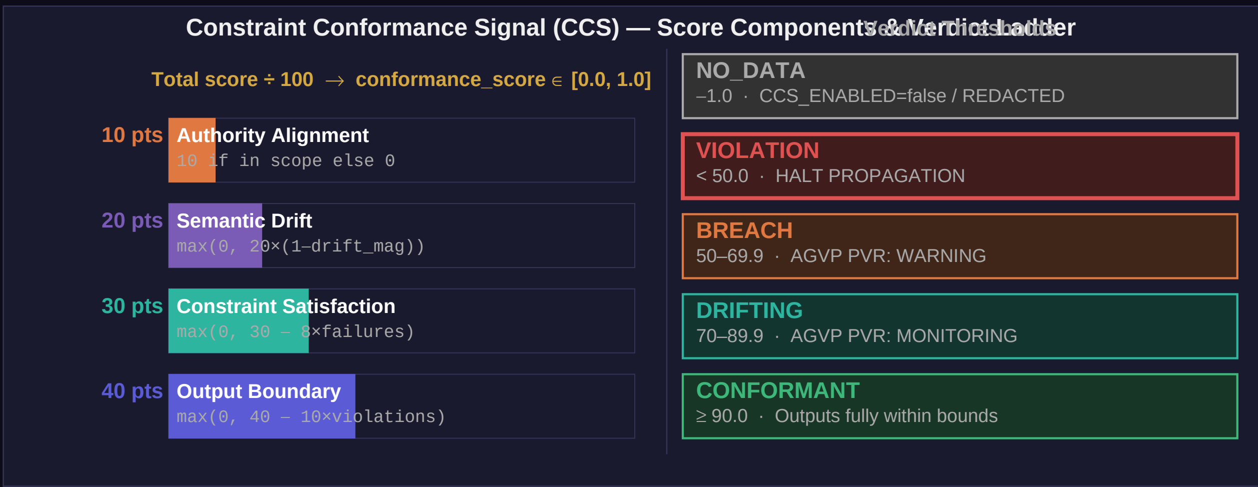

Figure 4: CCS score breakdown (left) and verdict ladder (right). Four components sum to a maximum of 100 points, normalized to [0.0, 1.0]. Each verdict threshold triggers a specific AGVP or session governance action.

3.2 Score Formula

```
OBS = max(0, 40 - 10 * boundary_violation_count) # output domain / prohibited classes
CSS = max(0, 30 - 8 * constraint_failure_count) # explicit constraint failures
SDS = max(0, 20 * (1 - drift_magnitude)) # cosine distance from authorized profile
AAS = 10 if authority_scope_satisfied else 0 # capability/resource scope check

ccs_score = OBS + CSS + SDS + AAS # range: [0, 100]
conformance_score = ccs_score / 100.0 # range: [0.0, 1.0] (BEV-INV-006)
drift_delta = 1.0 - conformance_score # per-turn drift contribution
cumulative_drift += drift_delta # session accumulator (BEV-INV-017)
```

3.3 AGVP Integration

The CCS-AGVP integration completes the AGVP input space. Structural signals (CES, fragmentation) were covered by RFC-ATF-4. Semantic drift (DSPP) was covered by RFC-ATF-4. Behavioral output conformance is now covered by CCS:

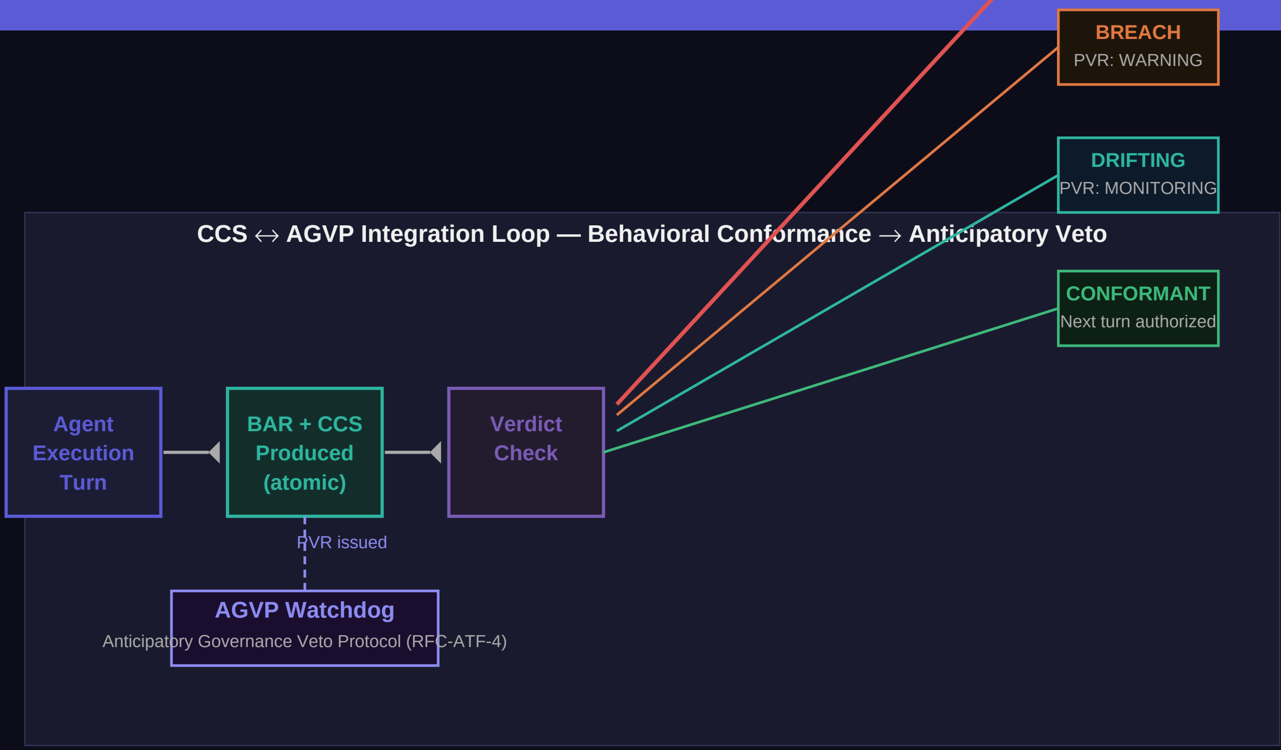

Figure 5: CCS-AGVP integration loop. Verdicts DRIFTING, BREACH, and VIOLATION each trigger escalating AGVP responses. VIOLATION triggers HALT propagation — the session is terminated before further outputs accumulate.

3.4 CCS Invariants

BEV-INV-005	**Mandatory CCS per BAR (atomic)**
Statement:	Every BAR MUST be accompanied by a CCS computation in the same atomic operation. No valid BAR without CCS.
Formal:	`EXISTS(CCS c) WHERE c.bar_id = b.bar_id AND c.created_atomically_with(b)`

BEV-INV-006	**CCS Conformance Score Bounds**
Statement:	conformance_score MUST be in [0.0, 1.0]. drift_delta = 1.0 - conformance_score. All components non-negative.
Formal:	`0.0 <= c.conformance_score <= 1.0 AND c.drift_delta = 1.0 - c.conformance_score`

BEV-INV-007	**CRITICAL/DRIFTING Verdict Triggers AGVP**
Statement:	verdict DRIFTING or worse MUST set watchdog_triggered=true and MUST trigger PVR issuance to AGVP watchdog.
Formal:	`c.verdict IN {DRIFTING,BREACH,VIOLATION} IMPLIES c.watchdog_triggered=True AND EXISTS(PVR p)`

BEV-INV-008	**Cumulative Drift Triggers HALT**
Statement:	When cumulative_drift > DRIFT_THRESHOLD (default 0.35), verdict MUST be HALT. THRESHOLD max 0.50 in production.
Formal:	`cumulative_drift > DRIFT_THRESHOLD IMPLIES verdict=HALT AND no subsequent turn permitted`

BEV-INV-009	**CCS History Append-Only**
Statement:	CCS records are append-only, chain-linked to CTCHC. No UPDATE or DELETE on atf_constraint_conformance_signals.
Formal:	`CCS records ordered by turn_index are strictly monotone AND chain_link_hash matches CTCHC`

BEV-INV-017	**Drift Accumulator Isolated per Session**
Statement:	cumulative_drift is independent per session_id. On process restart, MUST reload from last persisted CCS record.
Formal:	`cumulative_drift(S_1) is independent of cumulative_drift(S_2) for all S_1 != S_2`

4. Cross-Turn Coherence Hash Chain (CTCHC)

The CTCHC provides session-level behavioral integrity for multi-turn agent sessions. It chains every BAR's output hash with the prior link and governing receipt ID, seeding from a genesis hash bound to the session identity. The resulting sealed CTCHC record is the Session Integrity Proof (SIP) — offline verifiable without any OMNIX infrastructure.

4.1 Chain Construction

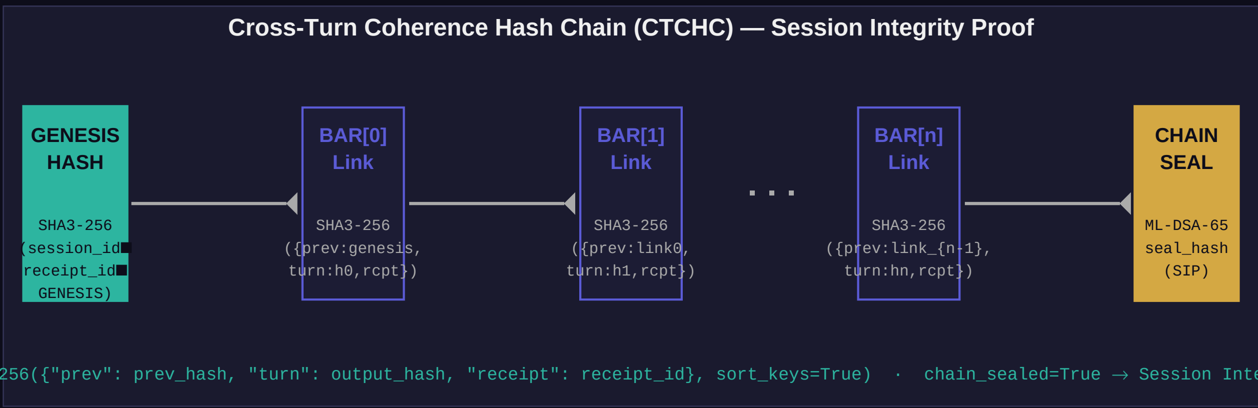

Figure 6: CTCHC hash chain. The genesis hash binds the chain to session_id and governing_receipt_id. Each link encodes the previous link, current turn's output hash, and receipt. The ML-DSA-65 chain seal over the final hash is the Session Integrity Proof.

4.2 Chain Link Formula

```
# Genesis (turn_index = 0):
genesis_hash = SHA3-256(
session_id + '||' + governing_receipt_id + '||' + 'OMNIX-CTCHC-GENESIS'
)

# Per-turn link (n >= 0):
link[n] = SHA3-256(
json.dumps({
'prev': genesis_hash if n == 0 else link[n-1],
'turn': SHA3-256(output_text.encode('utf-8')), # output_hash
'receipt': governing_receipt_id,
}, sort_keys=True).encode()
)

# Chain seal at session close:
seal_hash = SHA3-256(genesis_hash || all_link_hashes || final_chain_hash)
ctchc_seal = base64.b64encode(dilithium3.sign(seal_hash.encode(), platform_secret_key))
```

4.3 Offline Verification (6 steps)

Step	Check	Failure meaning
1	ctchc_seal over seal_hash verifies with ML-DSA-65 public key	CTCHC record tampered
2	len(all_bar_ids) = turn_count, turn_index[0..N-1] complete	Missing turns

Step	Check	Failure meaning
3	Each BAR passes individual content_hash + pqc_signature check	BAR tampered
4	Recompute genesis_hash from BAR_0 fields, compare to CTCHC.genesis_hash	Session identity tampered
5	Reconstruct all link[n], compare to BAR_n.chain_link	Turn substituted or reordered
6	chain_link(N-1) = CTCHC.final_chain_hash	Chain sealed over different sequence

4.4 CTCHC Invariants

BEV-INV-010	**Chain Initialized Before First BAR**
Statement:	CTCHC genesis MUST be created before BAR_0. genesis_hash = SHA3-256(session_id||'|'||receipt_id||'|'||'OMNIX-CTCHC-GENESIS').
Formal:	`EXISTS(CTCHC C) WHERE C.session_id = S.session_id AND C.initialized_before_BAR_0`

BEV-INV-011	**Chain Link Hash Construction**
Statement:	link[n] = SHA3-256(json.dumps({'prev': prev_hash, 'turn': output_hash, 'receipt': receipt_id}, sort_keys=True)). Deterministic.
Formal:	`L_n.chain_link_hash = SHA3-256(canonical({prev: L_{n-1}.hash, turn: output_hash_n, receipt}))`

BEV-INV-012	**Gaps in Turn Sequence Fail Verification**
Statement:	Verification MUST fail if any turn_index in [0, N-1] is absent. A sealed CTCHC claiming N turns but N-1 links is rejected.
Formal:	`{L.turn_index : L in C.links} = {0, 1, ..., N-1} OR verify(C) = FAIL`

BEV-INV-013	**Seal Covers Complete Chain**
Statement:	seal_hash = SHA3-256(genesis_hash || all_link_hashes || current_tip_hash). Partial seals invalid.
Formal:	`C.seal_hash = SHA3-256(C.genesis_hash || all_link_hashes || C.current_tip_hash) for all N links`

BEV-INV-014	**Seal ML-DSA-65 Signed Before OEP Export**
Statement:	CTCHC seal MUST be ML-DSA-65 signed before OEP export or regulatory submission. Unsigned seal not accepted.
Formal:	`dilithium3.verify(C.seal_hash.encode(), C.pqc_seal_signature, platform_public_key) = True`

BEV-INV-018	**Every Link's Receipt ID Matches Chain Receipt**
Statement:	Every chain link MUST carry the same governing_receipt_id as the CTCHC genesis. Prevents cross-session splicing.
Formal:	`For all L in C: L.governing_receipt_id = C.governing_receipt_id`

5. BEV Layer Composition

BAR, CCS, and CTCHC interact at each execution turn in a strictly ordered pipeline. The ordering guarantees that the CCS is embedded in the BAR before the content_hash is computed, and the chain_link from the prior turn is incorporated before the BAR is sealed.

5.1 Per-Turn Execution Timeline

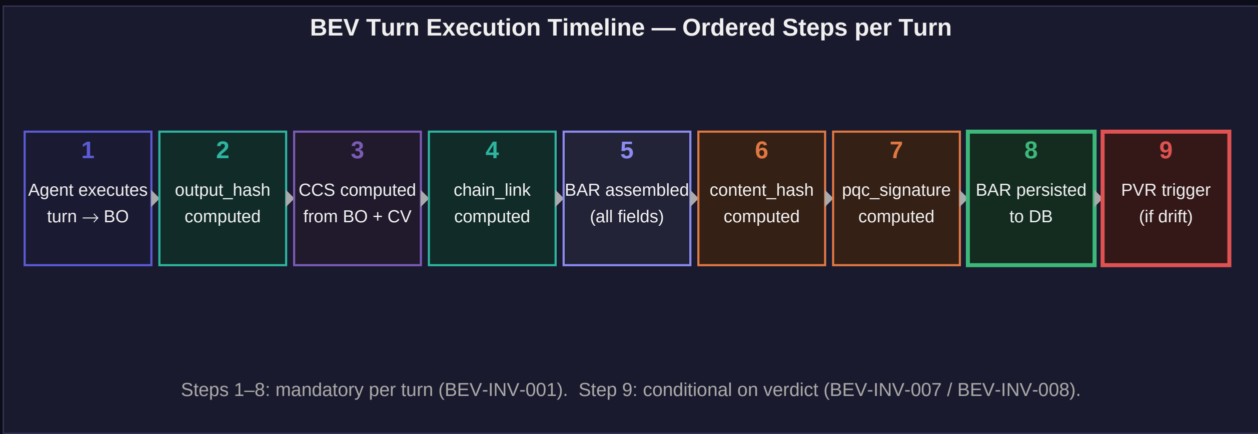

Figure 7: BEV per-turn execution timeline. Steps 1–8 are mandatory for every turn (BEV-INV-001). Step 9 (AGVP trigger) is conditional on the CCS verdict. Session close adds steps 12–17 for CTCHC sealing.

5.2 Cross-Layer Integration Points

BEV layer	Integrates with	Integration point
BAR	Governing Receipt (L1-L2)	governing_receipt_id in every BAR; queryable from any ATF receipt
CCS	AGVP (L4 RFC-ATF-4)	DRIFTING/BREACH/VIOLATION trigger PVR issuance via AGVP watchdog
CTCHC	OEP (L3 RFC-ATF-3)	CTCHC + BAR records included in OEP forensic packages
BAR	TCS (L5 RFC-ATF-5)	bar_timestamp_ns anchors to Temporal Context Snapshot for 7yr retention

6. Compliance Designation — ATF-BEV-Compliant

An implementation is designated **ATF-BEV-Compliant** — the sixth and highest compliance tier — when it satisfies all requirements of RFC-ATF-1 through RFC-ATF-6: ATF-CGL-Compliant (baseline) plus BAR operational, CCS operational, CTCHC operational, and BEV formal verification passing.

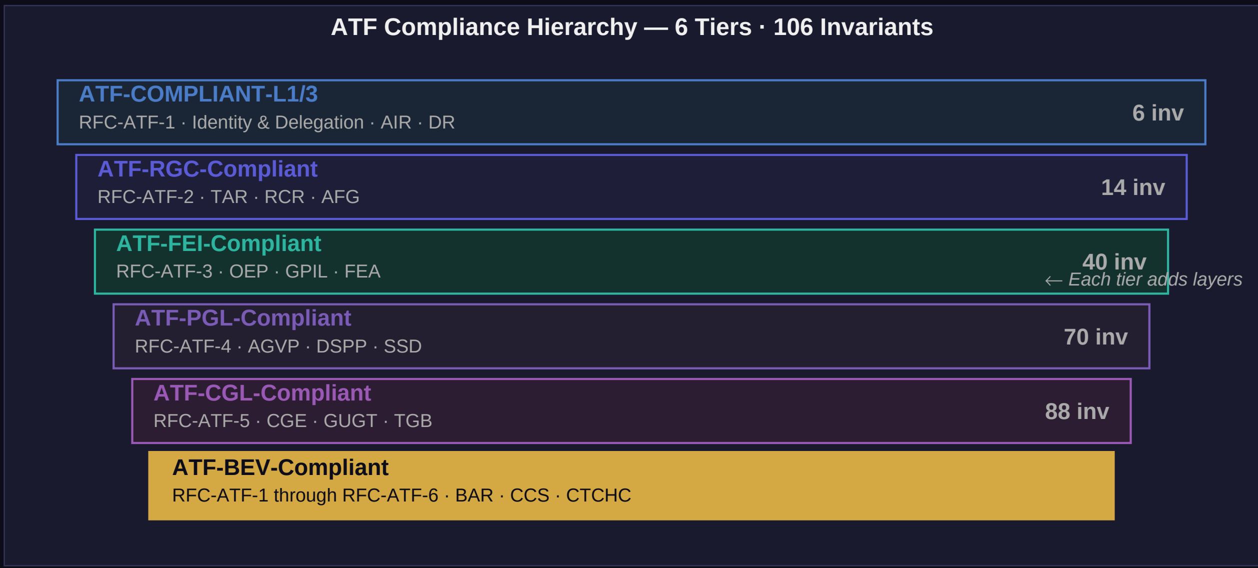

Figure 8: ATF compliance hierarchy. ATF-BEV-Compliant (gold, top) requires compliance with all 6 RFCs and 106 invariants. Each tier is additive — BEV does not supersede prior tiers; it extends them.

6.1 Compliance Requirements Summary

Requirement	Condition	BEV-INVs covered
ATF-CGL-Compliant baseline	All RFC-ATF-1–5 requirements satisfied	ATF-INV-001–006 + 82 prior
BAR operational	BEV_ENABLED=true, all turns produce persisted BARs	BEV-INV-001–004, 015–016
CCS operational	CCS_ENABLED=true, AGVP integration active for DRIFTING+	BEV-INV-005–009, 017
CTCHC operational	CTCHC_ENABLED=true, all completed sessions sealed	BEV-INV-010–014, 018
BEV formal verification	run_bev_proofs.py: all 5 targets unsat/structural	BEV-FVS-001–005

7. Combined Invariant Summary — RFC-ATF-6

RFC-ATF-6 introduces 18 new invariants across the three BEV protocol families. Combined with 88 invariants from RFC-ATF-1 through RFC-ATF-5, the ATF stack reaches 106 formally specified invariants.

Invariant	Family	Statement (summary)
BEV-INV-001	BAR	Mandatory BAR before output delivered
BEV-INV-002	BAR	content_hash = SHA3-256(output■receipt■index)
BEV-INV-003	BAR	HALTED BAR → immediate session halt
BEV-INV-004	BAR	ML-DSA-65 sealing + offline verifiability + append-only
BEV-INV-005	CCS	Mandatory CCS per BAR — atomic production
BEV-INV-006	CCS	conformance_score ∈ [0.0, 1.0]; drift derived
BEV-INV-007	CCS	DRIFTING or worse → AGVP watchdog triggered
BEV-INV-008	CCS	Cumulative drift > threshold → HALT
BEV-INV-009	CCS	CCS history append-only, hash-linked to CTCHC
BEV-INV-010	CTCHC	Chain initialized before first BAR
BEV-INV-011	CTCHC	link = SHA3-256({prev, turn, receipt})
BEV-INV-012	CTCHC	Gaps in turn sequence → verification fails
BEV-INV-013	CTCHC	Seal covers complete chain (genesis to tip)
BEV-INV-014	CTCHC	Seal ML-DSA-65 signed before OEP export
BEV-INV-015	BAR	Empty output_text → VIOLATION
BEV-INV-016	BAR	BAR id format: BAR-{HEX16}
BEV-INV-017	CCS	Drift accumulator isolated per session_id
BEV-INV-018	CTCHC	Every link's receipt_id MUST match chain receipt

7.1 Complete ATF Invariant Registry

Family	RFC	Count	Scope
ATF-INV	ATF-1	6	Identity & Delegation
RGC-INV	ATF-2	8	Runtime Continuity
GPIL-INV	ATF-3	3	Governance Policy Interop
ELR-INV	ATF-3	4	Evidence Lifecycle
EAP-INV	ATF-3	7	Evidence Archive Pipeline
OEP-INV	ATF-3	6	OMNIX Evidence Package
FEA-INV	ATF-3	5	Forensic Export Auth
FVP-INV	ATF-3	1	Forensic Verification Protocol
GECR-INV	ATF-3	6	Governance Execution Context Record

Family	RFC	Count	Scope
SGIP-INV	ATF-3	4	Semantic Governance Interop Protocol
DSPP-INV	ATF-4	7	Dynamic Semantic Portability
AGV-INV	ATF-4	6	Anticipatory Governance Veto
SSD-INV	ATF-4	3	Structural Shift Detection
FVS-INV	ATF-4	3	Formal Verification Suite
CGE-INV	ATF-5	7	Counterfactual Governance Engine
GUGT-INV	ATF-5	6	Grand Unified Governance Theory
TGB-INV	ATF-5	5	Temporal Governance Bridge
BEV-INV	ATF-6	18	Behavioral Execution Verification (BAR+CCS+CTCHC)
TOTAL	1–6	106	20 protocol families

8. Persistence Schema

8.1 atf_behavioral_anchor_records

```
CREATE TABLE IF NOT EXISTS atf_behavioral_anchor_records (
bar_id VARCHAR(64) PRIMARY KEY, -- BAR-{HEX16}
session_id VARCHAR(64) NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
agent_id VARCHAR(128) NOT NULL,
turn_index INTEGER NOT NULL,
output_hash VARCHAR(80) NOT NULL, -- sha256:64hex
output_hash_mode VARCHAR(16) NOT NULL, -- FULL|HASHED|REDACTED
output_payload JSONB, -- mode=FULL only
constraint_vector JSONB NOT NULL,
ccs_score NUMERIC(7,4),
ccs_verdict VARCHAR(16), -- CONFORMANT|DRIFTING|...
ccs_components JSONB,
chain_link VARCHAR(64) NOT NULL, -- 64 hex chars
genesis_hash VARCHAR(64), -- turn_index=0 only
session_start_ns BIGINT, -- turn_index=0 only
bar_timestamp_ns BIGINT NOT NULL,
issued_at TIMESTAMPTZ NOT NULL,
content_hash VARCHAR(64) NOT NULL,
pqc_signature TEXT NOT NULL, -- base64 ML-DSA-65
pqc_algorithm VARCHAR(16) NOT NULL DEFAULT 'ML-DSA-65',
atf_spec_version VARCHAR(8) NOT NULL DEFAULT '1.6',
bar_status VARCHAR(16) NOT NULL DEFAULT 'ACTIVE',
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

8.2 atf_constraint_conformance_signals

```
CREATE TABLE IF NOT EXISTS atf_constraint_conformance_signals (
ccs_id VARCHAR(64) PRIMARY KEY, -- CCS-{HEX16}
bar_id VARCHAR(64) REFERENCES atf_behavioral_anchor_records,
session_id VARCHAR(64) NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
turn_index INTEGER NOT NULL,
conformance_score NUMERIC(7,4) NOT NULL, -- [0.0, 1.0]
verdict VARCHAR(16) NOT NULL, -- CONFORMANT|DRIFTING|...
output_boundary_score NUMERIC(6,3),
constraint_satisfaction_score NUMERIC(6,3),
semantic_drift_score NUMERIC(6,3),
authority_alignment_score NUMERIC(6,3),
boundary_violation_count INTEGER,
constraint_failure_count INTEGER,
drift_magnitude NUMERIC(7,4),
```

```
cumulative_drift NUMERIC(7,4) NOT NULL,
drift_delta NUMERIC(7,4) NOT NULL,
watchdog_triggered BOOLEAN NOT NULL DEFAULT FALSE,
agvp_pvr_id VARCHAR(64),
halt_triggered BOOLEAN NOT NULL DEFAULT FALSE,
chain_link_hash VARCHAR(64),
computed_at_ns BIGINT NOT NULL,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

8.3 atf_coherence_hash_chains

```
CREATE TABLE IF NOT EXISTS atf_coherence_hash_chains (
ctchc_id VARCHAR(64) PRIMARY KEY, -- CTCHC-{HEX16}
session_id VARCHAR(64) UNIQUE NOT NULL,
governing_receipt_id VARCHAR(128) NOT NULL,
genesis_hash VARCHAR(64) NOT NULL,
final_chain_hash VARCHAR(64),
turn_count INTEGER NOT NULL DEFAULT 0,
all_bar_ids JSONB, -- ordered by turn_index
session_start_ns BIGINT,
session_close_ns BIGINT,
session_status VARCHAR(32) NOT NULL DEFAULT 'ACTIVE',
failure_turn_index INTEGER,
failure_reason TEXT,
seal_hash VARCHAR(64),
ctchc_seal TEXT, -- base64 ML-DSA-65
pqc_algorithm VARCHAR(16) NOT NULL DEFAULT 'ML-DSA-65',
chain_sealed BOOLEAN NOT NULL DEFAULT FALSE,
atf_spec_version VARCHAR(8) NOT NULL DEFAULT '1.6',
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

9. Formal Verification — OMNIX-FVS-1.0 BEV Extension

RFC-ATF-6 extends the OMNIX Formal Verification Suite (FVS-1.0, ADR-177) with five BEV proof targets. All proofs are Z3 SMT-checkable or demonstrable via structural collision resistance arguments.

Proof ID	Target	Method	Expected result
BEV-FVS-001	BAR binding completeness	Z3 boolean implication	unsat
BEV-FVS-002	Output hash structural uniqueness	SHA-256 collision resistance	structural
BEV-FVS-003	CCS score bounds [0.0, 1.0]	Z3 bounded real arithmetic	unsat
BEV-FVS-004	CCS component non-negativity	Z3 max(0,·) semantics	unsat
BEV-FVS-005	CTCHC chain monotonicity	SHA-256 structural induction	structural

9.1 BEV-FVS-003 — CCS Score Bounds (Z3)

```
from z3 import Real, Or, Solver, And, unsat
obs, css, sds, aas = [Real(x) for x in ['obs', 'css', 'sds', 'aas']]
s = Solver()
s.add(obs >= 0, obs <= 40)
s.add(css >= 0, css <= 30)
s.add(sds >= 0, sds <= 20)
s.add(And(Or(aas == 0, aas == 10)))
ccs_score = obs + css + sds + aas
# Claim: conformance_score = ccs_score/100 is always in [0.0, 1.0]
s.add(Or(ccs_score < 0, ccs_score > 100))
assert s.check() == unsat # No violation possible — BEV-INV-006 holds
```

Run all proofs: python omnix_core/formal_verification/run_bev_proofs.py

10. Regulatory Alignment

Regulation / Standard	Article / Section	BAR addresses	CCS addresses	CTCHC addresses
EU AI Act	Art. 9 — Risk management	Output attestation per turn	Continuous conformance monitoring	Session integrity proof
EU AI Act	Art. 12 — Record-keeping	Per-turn behavioral record	Conformance history projection	Complete session audit trail
EU AI Act	Art. 14 — Human oversight	Auditable output record	Drift signal for human review	Turn-by-turn oversight record
NIST AI RMF	MEASURE 2.6 — Monitoring	Attestation of actual outputs	Governance-native monitoring signal	Session coherence evidence
ISO/IEC 42001	§8.4 — Operational controls	PQC-signed behavioral record	Operational conformance per §8.4	Turn-by-turn control record
ISO/IEC 42001	§9.1 — Monitoring/evaluation	Verifiable output record	CCS is measurement artifact	Session history analysis
MiFID II	Art. 17 — Algo trading	Pre/post-output record	Conformance of trading outputs	Multi-turn session integrity
SOC 2 Type II	CC7.2 — Monitoring	Output monitoring evidence	Quantitative conformance evidence	Session integrity in audit

11. Novel Contributions

Contribution	Description	Prior art distinction
BAR — Behavioral Anchor Record	First cryptographic artifact binding agent outputs to governing receipts via PQC-signed content_hash = SHA3-256(output■receipt■index)	ML monitoring + output logging exist; PQC binding to governance receipt is new
CCS — Constraint Conformance Signal	First governance-native behavioral monitoring signal referencing actual CV from PQC-signed governing receipt	Arize/WhyLabs monitor against separately configured policies; CCS uses the receipt itself
CTCHC — Cross-Turn Coherence Hash Chain	Session-level hash chain seeded by governing receipt; SIP offline-verifiable without OMNIX infrastructure	Audit log hash chains exist; linkage to governance receipts for session-level forensic verification is new
CCS-AGVP Integration Loop	Behavioral conformance signal feeding directly into AGVP anticipatory veto protocol for PVR issuance	Completes AGVP input space; first behavioral-to-proactive-veto integration
Behavioral Attestation Chain	End-to-end verifiable chain: GR authorizes → BAR attests output → CCS measures conformance → CTCHC seals session	No prior standard closes authorization-to-behavioral-output chain
ATF-BEV-Compliant	Sixth compliance tier requiring 106 invariants across all ATF layers	Supersedes ATF-CGL-Compliant (RFC-ATF-5); highest and final ATF designation

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RFC-ATF-6 Version 1.0.0 — May 2026 — OMNIX QUANTUM LTD — Harold Nunes, Editor

Priority Records: OMNIX-PAR-2026-BAR-001 · OMNIX-PAR-2026-CCS-001 · OMNIX-PAR-2026-CTCHC-001